Employee Leave Management (ELM)

Fusion ERP | Phase 2 integration and improvement report
Team roll numbers: 24BCS046, 24BCS097 | Prepared: 26 October 2026

1. Executive summary

This report documents the ELM Phase 2 code submitted on the **elm-local-ai-trial** branch of ayu1223/Fusion-Integrated. The earlier minimal Phase 1 **modules/elm/** implementation was replaced with a more extensive **modules/leave/** implementation, accompanied by configuration, frontend, permission, and documentation changes. The submission demonstrates repository-level integration work; successful end-to-end operation has not been independently established.

2. Repository and version evidence

Phase 1 baseline: https://github.com/ayu1223/Fusion-Integrated/tree/phase1-elm-baseline/modules/elm

Final Phase 2 code: https://github.com/ayu1223/Fusion-Integrated/tree/elm-local-ai-trial/modules/leave

Existing pull request #1: https://github.com/ayu1223/Fusion-Integrated/pull/1

Git records reported in the user terminal: Phase 1 baseline **3318501**; integration commits **6213703** and **13e40bc**; Phase 2 integration **6efb1f0**; documentation reorganization **b5de311**. The last directly reported remote state for **elm-local-ai-trial** was **b5de311**; later path-correction commands were discussed but a resulting push was not shown.

The Phase 2 integration commit **6efb1f0** reported **45 changed files**, **6,735 insertions** and **123 deletions**. Those figures include prompt documentation and are not a measurement of tested functionality.

3. Key integration changes

- Replaced the earlier six tracked files under **modules/elm/** with the larger **modules/leave/** backend structure.
- Added leave-domain logic, Django models, service operations, API views and serializers, URL configuration, a migration file, and business-rule test code.
- Updated **config/settings/base.py** and **config/urls.py** to integrate the leave module into Fusion.
- Added **client/src/modules/leave/** pages, types, hooks, and routes, and updated client module registration.
- Updated **core/api/exceptions.py**, the permissions registry, and generated permission documentation.

4. Identified concerns and intended improvements

The supplied Phase 2 reference prompt material describes incorrect Django model choice definitions, references to a missing **created_at** field, missing or misconfigured API routes, and an absent shared **ValidationError** class. It also calls for verification of BR-EL-007 (vacation-leave conversion), BR-EL-011 (weekend handling), and BR-EL-012 (half-day overlap). These are useful traceability points, but reference-log descriptions alone do not prove the same faults were independently reproduced or fixed in the submitting environment.

5. Verification evidence and limitations

- **Verified locally through terminal output: git diff --check** produced no whitespace errors during the integration review; a search across config/core/modules/client returned no remaining **modules.elm** references in the queried paths.
- **Verified source-control operation:** the user pushed **b5de311** to **origin/elm-local-ai-trial** successfully.
- **Runtime blocker: python3 manage.py check** raised **ModuleNotFoundError: No module named django** because Django was missing from the active Python environment.

- **Not evidenced:** a completed Django system check, successful database migration, full frontend build, complete pytest suite, API role tests, or end-to-end user workflow on the submitting machine.

6. AI prompting records and provenance

Phase 2 prompt reference log: https://github.com/ayu1223/Fusion-Integrated/blob/elm-local-ai-trial/PHASE2_OLLAMA_PROMPTS1.md

Phase 2 prompt pack: https://github.com/ayu1223/Fusion-Integrated/blob/elm-local-ai-trial/OLLAMA_PROMPT_PACK.md

Phase 1 reconstructed prompts: https://github.com/ayu1223/Fusion-Integrated/blob/elm-local-ai-trial/phase1_LOG.md

Provenance qualification: The Phase 2 prompt documents were copied from a teammate/friend reference and later adapted to the submitting machine's home directory; the Phase 1 prompt document was reconstructed after original logs could not be located. They are supporting/reference documentation, **not authenticated raw interaction histories for the submitting pair**. Original execution logs should be added separately if required by the assignment.

7. Outstanding actions before deployment

- Activate or install the repository's intended Django Python environment, then run **manage.py check**.
- Run migrations with the project database available and execute the existing automated test suites.
- Build and launch the React client and Django server, then test leave submission, review, approvals, access control, and relevant business rules end-to-end.
- Confirm specification coverage and update the PR with reproducible command outputs or CI evidence.

8. Overall assessment

The Phase 2 branch represents a substantial structural and functional expansion over the minimal Phase 1 module. Repository changes and Git operations are documented, but deployment readiness is **not yet demonstrated**. Further environment setup, test execution, and expert review are recommended before production use.

Report basis: user-provided Git terminal outputs, committed-file list, supplied prompt documents, and linked repository/PR. Statements marked as unverified have not been inferred as successful outcomes.